

802_Quantum Computing Course - Session 1_Introduction & Program Overview

[week1-802-quantum computing.txt](#)

Quantum Computing Course - Session 1: Introduction & Program Overview

1. Session Agenda & Lecturer Introduction

First, we will discuss the program structure. On the first day during orientation, I was not able to walk you through the entire program structure. Now you have already gone through two courses as well. So they must have discussed their individual course outlines.

The plan for today is:

1. Discuss the entire program structure for your MSc.
2. Do a brief introduction to get to know you more (name, home country, background) to help plan future lessons.
3. Start the **Quantum Computing** introduction, focusing on a comparison between **conventional computing** versus **quantum computing**.
4. Depending on time, we may go into breakout rooms for tasks and discussion sharing.

We are starting at 10:40 now. We will go for 1 hour, then take a 10-minute break, and come back again. If I forget the break, please remind me via chat.

2. Master of Software Engineering Program Overview

2.1. PROGRAM HISTORY & CONTEXT

This is the first Master's program which Yoobee rolled out. We got approval in 2022 to start it. In 2023, we started in July with an initial intake of 11 students.

I joined Yoobee 3.5 years ago from India, where I was an Assistant Professor at the Manipal Institute of Technology (MIT Manipal). I was heavily involved in drafting and getting approval

for the Software Engineering program, so it is very close to my heart. I continue to teach to stay associated with the program.

Since 2023, student numbers have gradually increased from 10 to 20, then 30, and now it has stabilized. We are getting around 40 to 50 students per intake. In your class, there are 35 students right now, and it may go up to 40-45 in the next couple of weeks.

We are now getting seriously good students, especially since starting the MBI program in 2024 and the Masters of Management (MOM) program alongside your intake.

 **Student:** Yeah. Uh they gave a brief about it sir.  **Lecturer:** All right. Okay. So I will just walk you through the program structure.

2.2. PROGRAM STRUCTURE & TIMELINE

- It's a 180-credit entire program with three trimesters.
- **Trimester 1:** Runs from April till July.
- **Trimester 2:** Runs from July to November (with a week break after Trimester 1).
- **Trimester 3:** Starts in January and goes till April. There is a long break from November till December/January (4-5 weeks).

Trimester 1 Courses (60 credits total)

You have already gone through:

- **Professional Software Engineering (30 credits):** You have back-to-back classes on Saturday and Sunday.
- **Research Methods (15 credits)**
- **Quantum Computing (15 credits)**

Trimester 2 Courses (60 credits total)

There will be four core courses (each 15 credits, one class per week):

- Data Analytics (taught by Muhammad)
- Blockchain and Cloud Security (taught by Arshad)
- Intelligent Transportation System (taught by Mahesh)

Trimester 3 (60 credits total)

This is entirely project-based. You have two choices:

1. Internship

2. In-house Project

- **Development-based project:** Any software development project.
- **Research-based project.**

For in-house projects, we will provide an internal supervisor. For an external internship, there will be a host supervisor from the organization, who must be a technical person as per our requirement for this MSc.

 **Student:** I have a question like now in third trimester uh it's either the project or uh going in as an intern, right? Uh would it be okay if we apply and start an internship as of now?  **Lecturer:** Yes. Yes, you can do that. But that will not be counted for the 60 credit. Because you are a full-time student. So something which you are doing right now even as an intern that doesn't fall under the scope of the 60 credit. This 60-credit internship should be planned from November onwards till April. The minimum requirement is 400 hours within that period (November to 4th of April). It could be paid or unpaid. Work done before that in your personal capacity is entirely up to you.

 **Student:** Okay. And uh if we are doing a research or like a project like it's going to be something like where we identify uh a problem to any organization and we give them a solution by creating them a software or e-commerce side or whichever it is.  **Lecturer:** Mhm. Yes.

 **Student:** Okay.  **Lecturer:** Now assessment...

 **Student:** I have one. Yes. Thank you. So I have one question. Um if like uh the internship will start uh on the third trimester. So if a student start earlier, so can that student do like 40-hour internship alongside with full-time study?  **Lecturer:** Yeah. So in your last trimester, during holiday time, you can do full-time work. When it comes to an internship for the 60 credits (from the start of Trimester 3 till 4th of April 2027), if you are doing a 40-hour internship for the course, that is part of your studies and does not count towards your visa work limit. You still have the right to work an additional 25 hours per week. So you could work a maximum of 65 hours (40+25). But before the third trimester, if you are doing an internship, you have to comply with your visa regulations (max 25 hours during term time).

 **Student:** Okay. But before that, but before that if you are doing an internship then you have to comply with your visa regulations. So that is if it is not a holiday time you are supposed to work only for 25 hours a five hours internship. Yeah, got it. Okay, thank you.

 **Lecturer:** All right. Uh any other question regarding this?

2.3. ASSESSMENT OVERVIEW & DEADLINES

- **Professional Software Engineering** and the **Capstone Project** have three assessments each.
- All other courses have only two assessments.
- For my (**Quantum Computing**) course:
 - **Assessment 1 (Individual):** Due by Week 10. Weighted 40%.
 - **Assessment 2 (Group Project):** Due by Week 16. I will give a one-week grace period after Week 15, during the break time.
- I will release the first assessment in Week 3 and the last assessment in Week 7 or 8 so you have fair time to work on it.
- I will update exact submission dates on Blackboard.

We are using an 11-point scale. The passing criteria is 50+ (C- is the minimum grade).

2.4. RESUBMISSION POLICY (CRITICAL INFORMATION)

There are specific attempt limits for your program.

- **No more than two assessments** can be granted resubmissions or a reset opportunity **across the whole program** (across all ~28-29 assessment components).
- Resubmission/reset is for when you fail an assessment. To pass a course, you must score at least a C- grade in **both** assessments. If you fail one component, you fail the course, even if your aggregate is above C-.
- For your intake, this strict rule of only two resubmissions across Levels 8 and 9 is applicable (though it may be relaxed for future intakes).
- **If you resubmit, your grade is capped at C- (50)**, even if you score 80.
- **Recommendation:** If you have difficulty and think you may fail, seek an extension **a week or two before** the deadline with a valid reason. I will be liberal with MSE students to grant extensions to help you pass on time.

 **Lecturer:** Um any question on on this?

3. Quantum Computing Course Introduction

3.1. LEARNING OUTCOMES (LOs)

After studying this course, you should be able to:

1. Critically evaluate **quantum information protocols** in software engineering using the principle of mathematical structure of quantum mechanics to validate performance claim. (In essence, learn some **quantum algorithms**).
2. Design **quantum circuits** for those algorithms to run on quantum computers using software engineering principles.
3. Critically analyze the historical development of quantum and classical computation and its application to differentiate computational capabilities in software engineering. (We will emphasize LO3 today).
4. Communicate the application and critical analysis of quantum concepts to diverse software industry audiences (e.g., through reports).

3.2. ASSESSMENT DETAILS FOR QUANTUM COMPUTING

- **Assessment 1:** Individual theory assignment. Weight: 40%. Will be discussed in Week 3.
- **Assessment 2:** Group project. Maximum of **two students per group**. You must form **different groups** from those in Muhammad's (Professional Software Engineering) course to diversify your network and learn from different people.

3.3. BLACKBOARD, SUBMISSION & TURNITIN

- Check if you can access the "April 2026 intake" class and the assessments on Blackboard.
- Submissions will be made via the "Assessment Submission" area on Blackboard.
- **Turnitin:** I have created a class for you on Turnitin and shared the Class ID and Enrollment Key in the chat. Please join it.
 - For the first assessment, you must submit through Turnitin for a similarity check.
 - Similarity needs to be **less than 20%**. You can submit up to three times to check and polish your work.
 - We can also see AI scores. When I release the assessment, I will be precise about what you can/cannot use AI for, and any AI use must be acknowledged.

4. Student Introductions

Lecturer asks students to briefly introduce themselves (name, country, background). A list of student introductions follows in the transcript, which has been condensed here for the summary.

5. Break & Start of Quantum Computing Content

 **Lecturer:** It's 11:32. Before we will start I think it's best we can take a 10 minutes break right itself instead of starting it and then taking the break. We'll come back by 10:40 straight. Okay.

[Break taken. Session resumes.]

 **Lecturer:** Now before I will start I want to start with your thoughts around quantum computing. What's your initial thoughts? What dramatically will it do that our conventional computing can't do?

 **Student:** Computing is like a estimate approximate something like that.  **Lecturer:** In layman's term.

 **Student:** Any activity that is done by computers.  **Lecturer:** Okay, that's good.

 **Student:** Processing.  **Lecturer:** All right. So, do you think is there anything which computers can't do? Is there anything in terms of a problem set which computers can't do?

 **Student:** Process emotions.  **Student:** Observing someone's human behavior.  **Lecturer:** But that's also through AI more or less people are doing it.

 **Student:** I think computer can do like anything except the speed... because it only consider zero and one.  **Lecturer:** Okay. So no but seriously there are certain set of problems which our computers can't solve... NP-hard problems. There is an algorithm but it may take ages to solve. Our entire finance industry's encryption is dependent on one such problem: **factorization**.

5.1. THE FACTORIZATION PROBLEM & SHOR'S ALGORITHM

- Example: Factor a large (e.g., 100-digit) number C into $A * B$. A brute-force approach would take ages, even on a supercomputer.
- **Shor's algorithm** is a **quantum algorithm** designed for quantum computers that can solve this factorization problem in seconds.
- Whoever masters and builds a operational quantum computer first will have a massive strategic advantage for the next 100-200 years, impacting geopolitics.
- **Task:** Figure out the current status of quantum hardware.

5.2. CONVENTIONAL COMPUTING STACK - A LAYERED REVIEW

Today's focus is comparing the computing stacks. We'll review the conventional stack (top-down) to understand the quantum stack (which we'll discuss bottom-up).

The Conventional Computing Stack:

1. **Problem:** Any real-world problem to solve.
2. **Algorithm:** Must be *deterministic* (because it's based on deterministic conventional physics).
3. **Software/Program:** Developed by software engineers based on the algorithm.
4. **System Software (OS, Compiler, etc.):** Manages hardware resources and provides an interface.
5. **Instruction Set Architecture (ISA):** The language of the processor (e.g., x86, ARM). Provided by the chip manufacturer to compiler designers.
6. **Microarchitecture:** How the ISA is implemented on the chip (pipelining, caching). Affects speed and cost.
7. **Logic & Circuits:** Built from logic gates (AND, OR, NAND). The NAND gate is functionally complete.
8. **Devices:** Transistors, capacitors. Built from semiconductors.
9. **Physics (Semiconductors):** The fundamental layer. Based on **conventional physics**.
 - Mastery of controlling electron movement through materials.
 - **Semiconductors** (like Silicon) are key because their conductivity can be controlled (via **doping** to create N-type or P-type).
 - This allows creation of perfect switches (transistors) that can represent 0 (no conductivity) and 1 (conductivity).
 - The entire computational stack is an abstraction built on top of this mastery of electron control.

Key Takeaway: When using a conventional computer, you are ultimately controlling the movement of electrons. Each layer of the stack abstracts away the complexity of the layer below.

5.3. LIMITATIONS OF CONVENTIONAL COMPUTING & THE ROAD AHEAD

- Because conventional computing is based on deterministic physics, it can only solve problems that have deterministic algorithms.
- Some problems (NP-hard) are intractable for these machines.
- Progress (like AI boom) happens when hardware advances to support solving larger instances of these problems.
- **Quantum computing** aims to use **quantum physics** to overcome these fundamental limitations.

5.4. CLASSICAL PHYSICS VS. QUANTUM PHYSICS

Aspect	Classical Physics (Conventional Computing)	Quantum Physics (Quantum Computing)
Determinism	Deterministic. Predictable outcomes.	Probabilistic. Outcomes associated with probabilities.
Variables	Continuous. Can take any value in a range.	Discrete (Quantized). Specific, separate levels/states.
Superposition	No superposition. An object is in one state at a time.	Superposition allowed. A system can be in multiple states simultaneously until measured.
Entanglement	No entanglement. Objects are independent.	Entanglement possible. Particles can be linked, instantaneously affecting each other.

Illustration of Quantum Behavior (Electrons):

- An electron in an atom has quantized energy levels.
- To move from one orbit to another, it requires an exact discrete amount of energy (a "quantum"). Providing less energy has no effect.
- This is contrary to classical intuition where partial energy would cause a partial movement.

5.5. FROM BITS TO QUBITS

- **Bit:** Building block of conventional computing. Can be **0** OR **1**.
- **Qubit:** Building block of quantum computing. As a concept, it can be **0 AND 1** simultaneously (superposition).
- Before measurement, a qubit exists in a superposition state (e.g., 70% probability of being 1, 30% probability of being 0). The act of measurement "collapses" it to a definite state (0 or 1).
- The hardware challenge is to master the control of qubits to achieve desired superposition states for computation.

5.6. COURSE PRACTICALITIES & THE FUTURE

- **Career:** Currently, quantum computing careers are mostly in fundamental research (PhD level). This course trains you for the upper layers of the future quantum stack (software, algorithms, frameworks).
- **Frameworks to Install for Next Class:**
 - **Qiskit (IBM):** Python-based Quantum Information Science Kit.
 - **Cirq (Google):** Python-based framework.
 - Install them and create an IBM account if possible.

- We will use simulators (like "QruDevice" from Sydney University) to run quantum circuits, as simulating quantum physics requires immense classical computing power.

5.7. PRE-CLASS TASK & YOUNG'S DOUBLE-SLIT EXPERIMENT

Tasks before next class:

1. Watch the shared video on **Young's Double-Slit Experiment**.
2. Research: How are qubits implemented in hardware? What are the current hardware approaches?
3. Install Qiskit and Cirq.
4. Revise the conventional computing stack.

Young's Double-Slit Experiment (Video Summary):

- Historical debate: Is light a particle (Newton) or a wave (Huygens)?
- Young's experiment (1803): Shine light through two narrow slits.
- **Particle expectation:** Two bright lines on the screen.
- **Actual result:** An interference pattern of multiple bright and dark bands.
- **Explanation:** Light behaves as a **wave**. Waves from the two slits interfere constructively (bright bands) and destructively (dark bands).
- This proved light's wave nature. Combined with Einstein's photoelectric effect (proving particle nature), it led to the concept of **wave-particle duality**—a core idea resonating with quantum superposition.

 **Lecturer:** Isn't it amazing? Like you all were expecting that there will be two bright spots... This was the first experiment which proved that the light is a wave as well... The kind of settled principle is wave particle duality and that kind of align well with superpositioning effect.

6. Summary & Master Level Resources

6.1. TRANSACTIONAL NOTIFICATIONS SUMMARY

- **Program Structure:** 180 credits, 3 trimesters (Apr-Jul, Jul-Nov, Jan-Apr). Trimester 3 is a 60-credit capstone (internship or project).
- **Trimester 3 Internship:** 400-hour minimum between Nov and 4th Apr. Can be paid/unpaid. Separate from personal/visa-limited work.

- **Assessment Deadlines (Quantum Computing):** Assessment 1 (Individual, 40%) - Week 10. Assessment 2 (Group of 2, different from other courses) - Week 16.
- **Critical Policy:** Maximum of **two** resubmissions/resets across the **entire program**. Resubmission grades are capped at C- (50). Seek extensions if struggling.
- **Platforms:** Check Blackboard access. Join the Turnitin class (ID & key shared). Similarity must be <20%.
- **Next Class:** On campus (room TBC if >35 students). Install **Qiskit** and **Cirq** frameworks.

6.2. CORE ACADEMIC KNOWLEDGE POINTS

- **Motivation for QC:** Solves classically intractable problems (e.g., factorization via **Shor's algorithm**), threatening current encryption.
- **Conventional Stack:** Abstraction layers from Problem -> Algorithm -> Software -> System Software -> ISA -> Microarchitecture -> Logic -> Devices -> Semiconductor Physics. Fundamentally about controlling electron flow.
- **Physics Foundations:** Conventional computing relies on **deterministic, continuous-variable** classical physics. Quantum computing leverages **probabilistic, discrete, superposition- and entanglement-enabled** quantum physics.
- **Qubit vs Bit:** Bit is 0 **or** 1. Qubit (conceptually) can be 0 **and** 1 (superposition) until measured, enabling parallel computation.
- **Wave-Particle Duality:** Exemplified by Young's double-slit experiment (wave interference) and the photoelectric effect (particle nature). Underpins the quantum superposition concept.

6.3. RECOMMENDED LEVEL 9 LEARNING RESOURCES

- **Textbooks:**
 - *Quantum Computing for Computer Scientists* by Noson S. Yanofsky and Mirco A. Mannucci. (Primary text mentioned by lecturer).
 - *Quantum Computation and Quantum Information* by Michael A. Nielsen and Isaac L. Chuang (The standard reference, often called "the bible" of QC).
 - *An Introduction to Quantum Computing* by Phillip Kaye, Raymond Laflamme, and Michele Mosca.
- **Frameworks & Official Docs:**
 - **Qiskit Documentation:** <https://qiskit.org/documentation/> (IBM's open-source SDK).
 - **Cirq Documentation:** <https://quantumai.google/cirq> (Google's framework).
 - **Microsoft Q# Documentation:** <https://docs.microsoft.com/en-us/quantum/> (Alternative framework ecosystem).
- **Academic Papers & Reviews:**

- "Quantum algorithms: an overview" by Montanaro, A. (2016). *npj Quantum Information* 2, 15023. A great review of major quantum algorithms.
 - "Quantum computational supremacy" by Arute, F. et al. (2019). *Nature* 574, 505–510. (Google's "quantum supremacy" experiment).
 - Look for recent papers in *Nature*, *Science*, *Physical Review Letters*, and *npj Quantum Information* on hardware progress (superconducting qubits, ion traps, topological qubits, etc.).
 - **Additional Learning:**
 - **IBM Quantum Experience:** <https://quantum-computing.ibm.com/> (Access real quantum processors and simulators).
 - **Michael Nielsen's "Quantum Country" Guide:** An innovative interactive essay on quantum mechanics and computation.
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中文全文翻译

量子计算课程 - 第1讲：介绍与项目概述

1. 课程议程与讲师介绍

首先，我们将讨论项目结构。在迎新日的第一天，我没能带大家完整了解整个项目结构。现在你们已经完成了两门课程，他们应该已经讨论过各自的课程大纲。

今天的计划是：

1. 讨论你们硕士项目的整体结构。
2. 进行简短介绍，以便更多地了解你们（姓名、国籍、背景），以帮助规划未来的课程。
3. 开始量子计算的介绍，重点是比较传统计算与量子计算。
4. 根据时间，我们可能会进入分组讨论室进行任务和分享讨论。

我们现在从10:40开始。我们将进行1小时，然后休息10分钟，再回来。如果我忘记休息，请通过聊天提醒我。

2. 软件工程硕士项目概述

2.1. 项目历史与背景

这是Yoobee推出的第一个硕士项目。我们在2022年获得批准启动。2023年7月，我们招收了首批11名学生。

我3年半前从印度加入Yoobee，之前在印度曼尼帕尔理工学院（MIT Manipal）担任助理教授。我深度参与了软件工程项目的起草和审批工作，因此对它感情深厚。我继续任教以保持与项目的联系。

自2023年以来，学生人数从10人逐渐增加到20人、30人，现已趋于稳定。我们每期招收约40至50名学生。你们班目前有35名学生，未来几周可能会增加到40-45人。

我们现在招到了非常优秀的学生，特别是在2024年启动MBI项目以及与你们同期启动管理硕士（MOM）项目之后。

 **学生：** 是的，老师。他们简要介绍过了。 **讲师：** 好的。那么我就带大家过一遍项目结构。

2.2. 项目结构与时间线

- 这是一个总计180学分的项目，包含三个学期。
- **第一学期：** 从4月到7月。
- **第二学期：** 从7月到11月（第一学期后有一周休息）。
- **第三学期：** 从1月开始到4月结束。11月到12月/1月之间有长假（4-5周）。

第一学期课程（总计60学分）

你们已经完成：

- **专业软件工程（30学分）：** 周六和周日有连续课程。
- **研究方法（15学分）**
- **量子计算（15学分）**

第二学期课程（总计60学分）

将有四门核心课程（每门15学分，每周一节课）：

- 数据分析（由Muhammad授课）
- 区块链与云安全（由Arshad授课）
- 智能交通系统（由Mahesh授课）

第三学期（总计60学分）

这完全是基于项目的。你有两个选择：

1. 实习

2. 校内项目

- **开发类项目：** 任何软件开发项目。
- **研究类项目。**

对于校内项目，我们将提供内部导师。对于外部实习，组织中会有一位主办方导师，根据我们对此硕士项目的要求，该导师必须是技术人员。

 **学生：** 我有个问题，在第三学期，要么做项目，要么去实习，对吗？如果我们现在申请并开始实习可以吗？
 **讲师：** 可以，你可以那样做。但那不会计入60学分。因为你是全日制学生。所以你现在做的任何实习工作都不属于这60学分的范围。这60学分的实习应该计划在11月到4月之间进行。最低要求是在此期间（11月到4月4日）完成400小时。可以是有偿或无偿的。在此之前以个人身份所做的工作完全由你自己决定。

 **学生：** 好的。如果我们做研究或项目，是不是类似于我们为某个组织识别问题，然后通过为他们创建软件或电子商务等解决方案来解决问题？
 **讲师：** 嗯，是的。

 **学生：** 好的。
 **讲师：** 现在说评估...

 **学生：** 我有一个问题。谢谢。我的问题是，如果实习在第三学期开始。那么如果学生提前开始，他能否在全日制学习的同时进行每周40小时的实习？
 **讲师：** 可以。在你的最后一个学期，假期期间，你可以全职工作。至于那60学分的实习（从第三学期开始到2027年4月4日），如果你为课程做每周40小时的实习，那是你学习的一部分，不计入签证工作限制。你仍然有权每周额外工作25小时。所以你最多可以工作65小时（40+25）。但在第三学期之前，如果你在做实习，你必须遵守签证规定（学期期间最多25小时）。

 **学生：** 好的。但在那之前，如果做实习，就必须遵守签证规定。也就是说，如果不是假期，你每周只能工作25小时。明白了。好的，谢谢。
 **讲师：** 好的。关于这个还有其他问题吗？

2.3. 评估概述与截止日期

- **专业软件工程和毕业设计项目**各有三项评估。
- 所有其他课程只有两项评估。
- 对于我的（量子计算）课程：
 - **评估1（个人）：** 第10周截止。占40%。
 - **评估2（小组项目）：** 第16周截止。第15周之后我会给一周的宽限期，在休息期间。

- 我将在第3周发布第一项评估，在第7或8周发布最后一项评估，以便你们有充足的时间完成。
- 我将在Blackboard上更新确切的提交日期。

我们使用11分制。及格标准是50分以上（C-是最低等级）。

2.4. 重交政策（重要信息）

你们的项目有特定的尝试次数限制。

- **在整个项目期间**（所有约28-29个评估组件中），**最多只有两次**评估可以被授予重交或重考机会。
- 重交/重考适用于评估不及格的情况。要通过一门课程，你必须在**两项**评估中都至少获得C-等级。如果你一项不及格，即使总分高于C-，该课程也不及格。
- 对于你们这届，这个严格的规则（在整个8级和9级只有两次重交机会）是适用的（尽管未来入学可能会放宽）。
- **如果你重交，你的成绩将被限制在C-（50分），即使你得了80分。**
- **建议：**如果你有困难并认为可能不及格，请在截止日期**前一两周**提出有效的理由申请延期。我会对MSE学生放宽，批准延期以帮助你们按时通过。

 讲师：关于这个有什么问题吗？

3. 量子计算课程介绍

3.1. 学习成果

学习本课程后，你应该能够：

1. 运用量子力学的数学结构原理，批判性地评估软件工程中的量子信息协议，以验证性能声明。（本质上，学习一些量子算法）。
2. 运用软件工程原理，为在量子计算机上运行这些算法设计量子电路。
3. 批判性地分析量子 and 经典计算的历史发展及其在区分软件工程中计算能力方面的应用。（今天我们将强调LO3）。
4. 向不同的软件行业受众传达量子概念的应用和批判性分析（例如，通过报告）。

3.2. 量子计算评估详情

- **评估1：** 个人理论作业。权重：40%。将在第3周讨论。

- **评估2：** 小组项目。**每组最多两名学生。**你必须组成与Muhammad（专业软件工程）课程不同的小组，以拓展你的人脉并向不同的人学习。

3.3. BLACKBOARD、提交与TURNITIN

- 检查你是否能访问Blackboard上的"April 2026 intake"班级和评估。
- 提交将通过Blackboard上的"Assessment Submission"区域进行。
- **Turnitin：** 我已经在Turnitin上为你们创建了一个班级，并在聊天中分享了班级ID和注册密钥。请加入。
 - 对于第一项评估，你必须通过Turnitin提交以进行相似性检查。
 - 相似性需要**低于20%**。你可以提交最多三次来检查和润色你的作业。
 - 我们也能看到AI分数。当我发布评估时，我会明确说明什么可以/不可以使用AI，任何AI使用都必须注明。

4. 学生自我介绍

讲师要求学生简要介绍自己（姓名、国家、背景）。转录稿中有一系列学生介绍，在此总结中已压缩。

5. 休息与量子计算内容开始

 **讲师：** 现在是11:32。在我们开始之前，我认为最好现在就休息10分钟，而不是开始后再休息。我们10:40准时回来。好的。

[休息。课程恢复。]

 **讲师：** 在开始之前，我想先听听你们对量子计算的看法。你们的初步想法是什么？它能做哪些我们的传统计算做不到的、具有变革性的事情？

 **学生：** 计算就像是估计、近似之类的。  **讲师：** 用通俗的话说。

 **学生：** 计算机所做的任何活动。  **讲师：** 好的，不错。

 **学生：** 处理。  **讲师：** 好的。那么，你们认为有什么是计算机做不到的吗？在问题集方面，有什么是这些计算机做不到的？

 **学生：** 处理情绪。  **学生：** 观察人的行为。  **讲师：** 但通过AI，人们或多或少也在做这些。

 **学生：** 我认为计算机几乎可以做任何事情，除了速度...因为它只考虑0和1。  **讲师：** 好的。但说真的，确实有一些问题集是我们的计算机无法解决的...NP难问题。存在算法，

但解决它可能需要很长时间。我们整个金融行业的加密就依赖于这样一个问题：因式分解。

5.1. 因式分解问题与Shor算法

- 例子：将一个大数（例如100位） C 分解为 $A * B$ 。暴力破解方法即使使用超级计算机也需要极长时间。
- **Shor算法**是一种为量子计算机设计的量子算法，可以在几秒钟内解决这个因式分解问题。
- 谁先掌握并建造出可运行的量子计算机，谁将在未来100-200年拥有巨大的战略优势，影响地缘政治。
- **任务：** 了解量子硬件的当前状况。

5.2. 传统计算栈 - 分层回顾

今天的重点是比较计算栈。我们将回顾传统栈（自上而下）以理解量子栈（我们将自下而上讨论）。

传统计算栈：

1. **问题：** 任何需要解决的实际问题。
2. **算法：** 必须是确定性的（因为它基于确定性的传统物理）。
3. **软件/程序：** 由软件工程师根据算法开发。
4. **系统软件（操作系统、编译器等）：** 管理硬件资源并提供接口。
5. **指令集架构（ISA）：** 处理器的语言（例如x86，ARM）。由芯片制造商提供给编译器设计者。
6. **微架构：** ISA如何在芯片上实现（流水线、缓存）。影响速度和成本。
7. **逻辑与电路：** 由逻辑门（与、或、与非）构建。与非门在功能上是完备的。
8. **器件：** 晶体管、电容器。由半导体材料制成。
9. **物理（半导体）：** 基础层。基于传统物理。
 - 掌握了通过材料控制电子流动的技术。
 - **半导体**（如硅）是关键，因为它们的导电性可以被控制（通过掺杂形成N型或P型）。
 - 这使得可以创建完美的开关（晶体管），可以表示0（不导电）和1（导电）。
 - 整个计算栈是建立在这种电子控制技术之上的抽象层。

关键点： 使用传统计算机时，你最终是在控制电子的移动。栈的每一层都抽象掉了下一层的复杂性。

5.3. 传统计算的局限性与未来之路

- 因为传统计算基于确定性物理，它只能解决具有确定性算法的问题。
- 一些问题（NP难问题）对这些机器来说是难以处理的。
- 进步（如AI热潮）发生在硬件发展到足以支持解决这些问题的更大实例时。
- **量子计算**旨在利用**量子物理**来克服这些根本性限制。

5.4. 经典物理 vs. 量子物理

方面	经典物理（传统计算）	量子物理（量子计算）
确定性	确定性的。结果可预测。	概率性的。结果与概率相关。
变量	连续的。可以取范围内的任何值。	离散的（量子化的）。特定的、分离的能级/状态。
叠加	无叠加。一个物体一次处于一种状态。	叠加 是允许的。系统在测量前可以同时处于多种状态。
纠缠	无纠缠。物体是独立的。	纠缠 是可能的。粒子可以相互关联，即时影响彼此。

量子行为示例（电子）：

- 原子中的电子具有量子化的能级。
- 要从一个轨道移动到另一个，需要精确的离散能量（一个"量子"）。提供更少的能量没有效果。
- 这与经典直觉相反，在经典直觉中，部分能量会导致部分移动。

5.5. 从比特到量子比特

- **比特**：传统计算的基本单元。可以是**0 或 1**。
- **量子比特**：量子计算的基本单元。作为一个概念，它可以同时是**0 和 1**（叠加态）。
- 在测量之前，量子比特存在于叠加态（例如，是1的概率为70%，是0的概率为30%）。测量的行为使其"坍缩"到一个确定的状态（0或1）。
- 硬件挑战在于掌握控制量子比特以达到计算所需的叠加态。

5.6. 课程实践与未来

- **职业**：目前，量子计算的职业大多在基础研究领域（博士水平）。本课程为你们培训未来量子栈的上层（软件、算法、框架）。
- ****下节课需安装的框架ogram**